

ZINNWALD GEORGENFE, GermanyWMO: **105820**Lat: **50.7314N**Lon: **13.7517E**Elev: **877**StdP: **91.23**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.9	-13.4	-20.0	0.7	-14.8	-16.7	1.0	-12.5	15.6	0.0	14.2	-0.7	4.5	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	23.2	16.3	21.5	15.6	20.0	14.8	17.4	21.9	16.4	20.3	15.4	19.0	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	12.3	19.6	14.6	11.6	18.4	13.7	10.9	17.3	52.4	21.9	49.1	20.3	46.1	19.0	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.9	10.6	-15.9	26.7	3.8	1.8	-18.6	28.0	-20.8	29.0	-22.9	30.0	-25.7	31.3		
(5)			-16.4	19.4	3.7	0.9	-19.1	20.0	-21.2	20.6	-23.3	21.1	-26.0	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-4.0	-4.4	-0.5	3.4	8.8	11.7	14.0	14.5	10.0	5.7	0.1	-2.3
	DBStd	7.78	4.97	5.06	4.18	4.10	3.85	3.77	3.57	3.35	3.18	3.94	3.70	3.98
	HDD10.0	2325	434	404	325	201	69	26	6	4	39	141	297	380
	HDD18.3	4946	693	637	583	448	294	199	139	126	251	391	547	638
	CDD10.0	433	0	0	0	3	33	78	131	142	38	8	0	0
	CDD18.3	13	0	0	0	0	0	1	6	6	0	0	0	0
	CDH23.3	47	0	0	0	0	0	3	18	26	0	0	0	0
	CDH26.7	3	0	0	0	0	0	0	1	2	0	0	0	0
Wind	WSAvg	5.4	6.3	6.1	6.0	5.0	4.8	4.6	4.3	4.1	4.8	6.0	6.1	6.3
Precipitation	PrecAvg	694	46	40	47	39	67	78	88	84	57	49	51	50
	PrecMax	996	90	73	157	79	133	162	159	228	128	100	107	106
	PrecMin	412	6	9	15	2	13	35	24	10	21	9	0	16
	PrecStd	114	23	18	28	18	32	31	34	47	28	23	25	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	9.4	12.6	15.9	19.5	23.4	25.6	26.2	20.2	17.1	11.6	9.5
		MCWB	5.1	5.4	8.1	10.0	12.9	16.1	18.2	18.1	14.6	13.2	7.1	6.2
	2%	DB	4.2	6.3	8.9	13.8	18.0	21.1	23.2	23.5	18.5	14.0	8.8	6.9
		MCWB	3.0	4.1	5.9	8.8	12.0	15.2	16.1	16.9	14.0	11.2	6.5	5.0
	5%	DB	3.3	3.5	6.6	11.6	16.6	19.3	21.7	21.5	16.9	12.3	7.0	5.0
		MCWB	2.3	2.0	4.5	7.6	11.3	14.2	15.5	15.9	13.1	10.1	5.7	3.7
	10%	DB	1.9	1.7	4.8	9.6	15.1	17.7	19.9	19.9	15.4	10.9	5.2	3.3
		MCWB	0.9	0.9	3.4	6.2	10.5	13.4	14.8	15.0	12.2	9.2	4.3	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	6.1	8.7	10.7	14.1	17.5	18.6	18.8	15.6	13.7	8.2	7.1
		MCDB	6.0	8.4	10.8	14.5	17.5	21.0	24.3	23.9	18.8	16.5	9.5	8.9
	2%	WB	3.6	4.2	6.1	9.3	12.9	16.3	17.3	17.6	14.6	11.9	7.1	5.4
		MCDB	4.0	6.2	8.3	13.0	16.9	19.5	21.7	22.6	17.6	13.5	8.4	6.6
	5%	WB	2.5	2.4	4.8	8.1	11.8	15.0	16.2	16.4	13.6	10.6	5.8	4.0
		MCDB	3.0	3.3	6.4	11.0	15.7	18.0	20.3	20.7	16.4	11.9	6.9	4.7
	10%	WB	1.0	0.8	3.5	6.6	10.8	13.9	15.2	15.4	12.5	9.5	4.3	2.6
		MCDB	1.6	1.5	4.8	9.2	14.5	17.1	19.0	19.1	15.0	10.7	5.3	3.2
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.3	4.7	6.4	7.1	6.8	7.2	7.1	6.0	5.2	4.3	3.7
		MCDBR	5.8	6.5	7.5	9.0	8.9	8.6	9.3	9.1	7.8	6.7	7.0	5.4
		MCWBR	5.1	5.1	5.4	5.8	5.5	5.3	5.4	5.3	5.1	4.6	5.5	4.4
	5% WB	MCDBR	5.2	5.9	7.1	8.5	8.1	7.9	8.5	8.6	7.5	6.0	6.4	5.1
		MCWBR	5.1	4.9	5.5	5.7	5.5	5.2	5.5	5.4	5.0	4.6	5.3	4.5
Clear-Sky Solar Irradiance	taub	0.287	0.309	0.349	0.389	0.390	0.386	0.400	0.395	0.361	0.337	0.312	0.282	
	taud	2.392	2.374	2.305	2.236	2.261	2.294	2.257	2.293	2.369	2.420	2.441	2.411	
	Ebn at Noon	751	826	848	850	865	870	849	835	827	779	713	700	
	Edh at Noon	61	81	105	126	128	125	128	118	98	77	59	52	
All-Sky Solar Radiation	RadAvg	0.79	1.55	2.53	4.03	4.86	5.24	4.97	4.33	3.10	1.82	0.86	0.60	
	RadStd	0.14	0.24	0.36	0.53	0.58	0.44	0.57	0.41	0.42	0.24	0.11	0.07	